1)
$$\int\left(3x^5+5\cos x+\frac{7}{x}-9e^x-\frac{1}{1+x^2}\right)dx=$$

$$=\frac{1}{2}x^6+5\sin x+7\ln x-9e^x+\operatorname{arcctg}x+C$$

2)

Poniższy przykład będzie na egzaminie, tylko jako niewłaściwa

$$\hookrightarrow \int\frac{x\sqrt[3]{x}+\sqrt[4]{x}}{x^2}dx=\int\frac{x\cdot x^{\frac{1}{3}}}{x^2}dx+\int\frac{x^{\frac{1}{4}}}{x^2}dx=$$

$$=\int x^{-\frac{2}{3}}dx+\int x^{-\frac{7}{4}}dx=\frac{1}{1-\frac{2}{3}}\cdot x^{\frac{1}{3}}+\frac{1}{1-\frac{7}{4}}x^{-\frac{3}{4}}+C=3\sqrt[3]{x}-\frac{4}{3}x^{-\frac{3}{4}}+C$$

3)
$$\int(\operatorname{tg}x+\operatorname{ctg}x)^2dx=\int\left(\frac{\sin x}{\cos x}+\frac{\cos x}{\sin x}\right)^2dx=\int\left(\frac{\sin^2x+\cos^2x}{\sin x\cos x}\right)^2dx=$$

$$=\int\frac{1}{\sin^2x\cdot\cos^2x}dx=\int\frac{4}{\sin^22x}dx=4\int\frac{1}{\sin^22x}dx=-4\operatorname{ctg}2x\cdot\frac{1}{2}=$$

$$=-2\operatorname{ctg}2x+C$$

4)
$$\int\left(\cos\frac{x}{2}-\sin\frac{x}{2}\right)^2dx=\int\left(\cos^2\frac{x}{2}-2\sin\frac{x}{2}\cos\frac{x}{2}+\sin^2\frac{x}{2}\right)dx=$$

$$=\int(1-\sin x)dx=\int1dx-\int\sin x\,dx=x+\cos x+C$$

5)
$$\int(5-\cos x)^4\sin x\,dx=\left\{\begin{array}{l}t=5-\cos x\\ dt=\sin x\,dx\end{array}\right\}=\int t^4dt=\frac{1}{5}t^5+C=\frac{1}{5}(5-\cos x)^5+C$$

6)
$$\int \frac{(1-\ln x)^3}{x}\,dx = \left\{\begin{array}{l} t=1-\ln x \\ dt=-\frac{1}{x}\,dx \\ dx=-x\,dt \end{array}\right\} = \int \frac{t^3}{x}\cdot(-x)\,dt = -\int t^3\,dt = -\frac{1}{4}t^4+C =$$
$$= -\frac{1}{4}(1-\ln x)^4 + C$$

7) Dobre na kolokwium
$$\int \sin 3\sqrt{x}\,dx = \left\{\begin{array}{l} t=3\sqrt{x} \\ dt=3\cdot\frac{1}{2\sqrt{x}}\,dx \\ dx=\frac{2\sqrt{x}}{3}\,dt \\ dx=\frac{2t}{9}\,dt \end{array}\right\} = \int \sin t\cdot\frac{2t}{9}\,dt = \frac{2}{9}\int \sin t\cdot t\,dt =$$

$$= \frac{2}{9}\int t\cdot\sin t\,dt = \left\{\begin{array}{ll} f=t & f'=1 \\ g'=\sin t & g=-\cos t \end{array}\right\} = \frac{2}{9}\left(-t\cos t+\int\cos t\,dt\right) =$$

$$= \frac{2}{9}(-t\cos t+\sin t) = -\frac{2}{9}t\cos t+\frac{2}{9}\sin t =$$
$$= -\frac{2\sqrt{x}}{3}\cdot\cos 3\sqrt{x}+\frac{2}{9}\sin 3\sqrt{x}+C$$

8)
$$\int \frac{\ln x}{\sqrt{x}}\,dx = \int\left(\ln x\cdot x^{-\frac{1}{2}}\right)dx = \left\{\begin{array}{ll} f=\ln x & f'=\frac{1}{x} \\ g'=\frac{1}{\sqrt{x}} & g=2\sqrt{x} \end{array}\right\} =$$

$$= 2\sqrt{x}\cdot\ln x - \int\left(\frac{1}{x}\cdot 2\sqrt{x}\right)dx = 2\sqrt{x}\ln x - 2\int\left(\frac{1}{\sqrt{x}}\right)dx =$$
$$= 2\sqrt{x}\ln x - 4\sqrt{x} + C$$

9) Buba ma wyprzedzenie (albo wendete) do Staszica? Na kolosa ten przykład

$$\int \frac{\operatorname{arctg} 2x}{x^2}\,dx = \int \left(\operatorname{arctg} 2x \cdot \frac{1}{x^2}\right) dx = \left\{ \begin{array}{ll} f = \operatorname{arctg} 2x & f' = \frac{2}{1+4x^2} \\ g' = \frac{1}{x^2} & g = -\frac{1}{x} \end{array} \right\} =$$

$$= -\frac{\operatorname{arctg} 2x}{x} + \int \left(\frac{2}{1+4x^2} \cdot \left(-\frac{1}{x}\right)\right) dx =$$

Kolos z całek

Zad. 1

2 całki nieoznaczone

Zad. 2

całka oznaczona

Zad. 3

całka niewłaściwa

10) $$\int x \cdot \operatorname{arcctg} x \, dx = \left\{ \begin{array}{ll} f = \operatorname{arcctg} x & f' = \frac{-1}{1+x^2} \\ g' = x & g = \frac{1}{2}x^2 \end{array} \right\} =$$

$$= \frac{1}{2}x^2 \operatorname{arcctg} x - \int \frac{x^2}{2} \cdot \left(\frac{-1}{1+x^2}\right) dx =$$

$$= \frac{1}{2}x^2 \operatorname{arcctg} x + \frac{1}{2}\int \frac{x^2}{1+x^2}\,dx =$$

$$= \frac{1}{2}x^2 \operatorname{arcctg} x + \frac{1}{2}\int \left(\frac{-1}{1+x^2} + 1\right) dx =$$

$$= \frac{1}{2}x^2 \operatorname{arcctg} x + \frac{1}{2}\int \frac{-1}{1+x^2}\,dx + \frac{1}{2}\int 1\,dx =$$

$$= \frac{1}{2}x^2 \operatorname{arcctg} x + \frac{1}{2}\operatorname{arcctg} x + \frac{1}{2}x + C$$

11)

$$\int \frac{\pi - \arcsin x}{\sqrt{1-x^2}}\,dx = \int (\pi - \arcsin x)\cdot \frac{1}{\sqrt{1-x^2}}\,dx = \left\{\begin{matrix} f = \pi - \arcsin x & f' = \frac{-1}{\sqrt{1-x^2}} \\ g' = \frac{1}{\sqrt{1-x^2}} & g = \arcsin x \end{matrix}\right\} =$$

$$= \overbrace{\arcsin x(\pi - \arcsin x)}^{A} + \int \frac{\arcsin x}{\sqrt{1-x^2}}\,dx =$$

$$= \left\{\begin{matrix} t = \arcsin x \\ dt = \frac{1}{\sqrt{1-x^2}}\,dx \\ dx = \sqrt{1-x^2}\,dt \end{matrix}\right\} = A + \int \frac{t}{\sqrt{1-x^2}}\cdot\sqrt{1-x^2}\,dt = A + \int t\,dt =$$

$$= A + \frac{1}{2}t^2 + C = A + \frac{1}{2}\arcsin^2 x + C$$

12)

$$\int x^3 \cdot e^{-2x^2}\,dx = \left\{\begin{matrix} t = -2x^2 \Rightarrow x^2 = -\frac{1}{2}t \\ dt = -4x\,dx \\ dx = \frac{-dt}{4x} \\ \int x^3 \cdot e^{-2x^2}\,dx = \\ = \int -\frac{1}{2}t\cdot\cancel{x}\cdot e^t \cdot \frac{-1}{4\cancel{x}}\,dt = \\ = \int \frac{1}{8} e^t\,dt \end{matrix}\right\} =$$

$$\int e^{x^2}\,dx = ?$$

$$\int e^{-x^2}\,dx = ?$$

$$= \frac{1}{8}\int t\cdot e^t\,dt = \left\{\begin{matrix} f = t & f' = 1 \\ g' = e^t & g = e^t \end{matrix}\right\} = \frac{1}{8}\left(t\cdot e^t - \int e^t\,dt\right) =$$

$$= \frac{1}{8}\left(t\cdot e^t - e^t + C\right)$$